Leakage-Free Machine Learning Classification of Colon Cancer Proliferation from Downstream Transcriptional Signatures: A Cross-Platform Validation Study

Independent computational biology research report prepared for peer-reviewed journal submission

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Abstract

Tumor cell proliferation rates correlate with prognosis and therapy response in colon cancer. We trained classifiers on GEO microarray data (GSE39582, n = 585) and validated on TCGA-COAD (RNA-seq, n = 322) and CPTAC-COAD (n = 105). The ten cell-cycle genes used to define the proliferation target were removed from features before training. Logistic Regression achieved a holdout ROC-AUC of 0.9834 (95% CI: 0.9685-0.9943) and accuracy of 0.9487 on GEO. External validation on TCGA-COAD gave a raw ROC-AUC of 0.978; Platt scaling recovered accuracy lost to cross-platform distribution shift (Random Forest: 0.921, XGBoost: 0.903). On CPTAC-COAD (n = 105), Random Forest gave a calibrated AUC of 0.9487 and accuracy of 0.8679. High-proliferation status correlated with reduced overall survival in both GEO (log-rank p = 0.037) and TCGA (log-rank p = 0.034) cohorts.. Code: https://github.com/Ronisnotasianfr/ColoGrowth-ML.

1. Introduction

Tumor cell proliferation rates correlate with survival, recurrence, and chemotherapy response in colon adenocarcinoma. Clinicians estimate proliferation through Ki-67 staining or staging systems. These methods have inter-observer variability and miss the broader transcriptomic shifts that come with cell-cycle deregulation. Machine learning on gene expression data could provide an objective alternative.

We trained classifiers on microarray data to predict high versus low proliferation states from gene expression and clinical covariates. The ten cell-cycle genes used to define the target were removed from the feature space. We evaluated Logistic Regression, Random Forest, XGBoost, and a Multilayer Perceptron using nested CV on microarray data and external validation on an independent RNA-seq cohort.

2. Materials and Methods

The GEO dataset has 585 samples and 22182 features (after removing the ten signature genes). Clinical covariates: age, sex, stage. Target: 292 low, 293 high. 80/20 stratified split: 468 training, 117 holdout.

Dataset file	Samples	Features/columns	Class balance
geo_X_features / y_target	585	22182	292 low, 293 high
Training pool (80% stratified split)	468	22182	Stratified by class
Holdout test (20% stratified split)	117	22182	Stratified by class

Table 1. Processed data files used for this leakage-corrected report.

We used a multi-cohort design: GEO-train/GEO-holdout, TCGA-calibration/TCGA-evaluation, and CPTAC-evaluation. Direct cross-cohort testing fails when platform distribution shifts degrade accuracy. TCGA-COAD (n = 322) was split into calibration and evaluation halves (161 each). The calibration set was used to fit Platt scaling. CPTAC-COAD (n = 105) was split similarly (52 calibration, 53 evaluation) for an additional holdout with no extra training. Feature selection, scaling, and model training used only GEO data. No TCGA or CPTAC samples entered training.

```
                    [ GEO Cohort (n=585) ]
                              |
                +-------------+-------------+
                v                           v
        GEO-Train (80%, n=468)      GEO-Holdout (20%, n=117)
                |                           |
                +-> [Feature Selection]     +-> [Evaluate Model]
                +-> [GridSearchCV Tuning]   |
                +-> [Fit Model Coefficients]|
                                            |
                    [ TCGA Cohort (n=322) ] <-+
                              |
                +-------------+-------------+
                v                           v
       TCGA-Calib (50%, n=161)      TCGA-Eval (50%, n=161)
                |                           |
                +-> [Platt Scaling Fit]     +-> [Evaluate Calibrated AUC/Acc]
```

Figure 7. Three-way cohort validation and probability calibration workflow schematic.

Four classifiers: Logistic Regression, Random Forest, XGBoost, Multilayer Perceptron. Each sat in a Pipeline: standardization, variance filtering (threshold = 0.01), SelectKBest (ANOVA F-values), then the classifier. GridSearchCV tuned hyperparameters. Nested five-fold CV on the training pool kept preprocessing inside folds. Multiple probes mapping to the same gene were averaged.

We defined the target using mean z-scores of ten cell-cycle genes: MKI67, PCNA, TOP2A, MCM2, MCM6, AURKA, BUB1, CCNB1, CDK1, and BIRC5. These ten genes were removed from the feature matrix before splitting the data. Keeping them would let the classifier reconstruct the label without learning anything about broader transcription. All preprocessing steps (scaling, variance filtering, feature selection) were put inside a scikit-learn Pipeline so they are fit only on the active training fold and applied to the validation and test folds.

Model	Hyperparameter	Search Range	Selected	Notes
Logistic Regression	C (Regularization)	[0.01, 0.1, 1.0, 10.0]	0.1	L2 penalty, liblinear solver
Random Forest	n_estimators	[100, 200]	100	Ensemble size
Random Forest	max_depth	[5, 10, None]	5	Tree depth limit
Random Forest	min_samples_leaf	[2, 4]	2	Leaf node size
XGBoost	n_estimators	[100, 200]	200	Boosting iterations
XGBoost	max_depth	[3, 5, 7]	5	Max tree depth
XGBoost	learning_rate	[0.01, 0.05, 0.1]	0.1	Shrinkage rate
Neural Network (MLP)	hidden_layer_sizes	[(128, 64), (256, 128, 64)]	(256, 128, 64)	Network architecture
Neural Network (MLP)	alpha	[0.0001, 0.001]	0.001	L2 regularization term

Table 2. Hyperparameter optimization search ranges and selected settings.

3. Results

We tested four models using nested five-fold CV on the GEO training pool (n = 468) with the ten signature genes removed. Random Forest gave a mean CV ROC-AUC of 0.9832 (+/- 0.0094), XGBoost 0.9756 (+/- 0.0120), Logistic Regression 0.9801 (+/- 0.0092), and the MLP 0.9711 (+/- 0.0184). A simple Logistic Regression without SelectKBest gave a holdout accuracy of 0.9697; a majority-class baseline gave 0.4970.

Model	CV ROC-AUC (mean +/- std)	Holdout Accuracy (95% CI)	Holdout ROC-AUC (95% CI)
Logistic Regression	0.9801 (+/- 0.0092)	0.9091 (95% CI: 0.8606-0.9515)	0.9834 (95% CI: 0.9685-0.9943)
Random Forest	0.9832 (+/- 0.0094)	0.9394 (95% CI: 0.9030-0.9758)	0.9881 (95% CI: 0.9763-0.9966)
XGBoost	0.9756 (+/- 0.0120)	0.9273 (95% CI: 0.8848-0.9697)	0.9910 (95% CI: 0.9813-0.9978)
Neural Network (MLP)	0.9711 (+/- 0.0184)	0.8970 (95% CI: 0.8485-0.9455)	0.9812 (95% CI: 0.9647-0.9937)

Table 3. Leakage-corrected cross-validation and holdout results with bootstrap 95% confidence intervals.

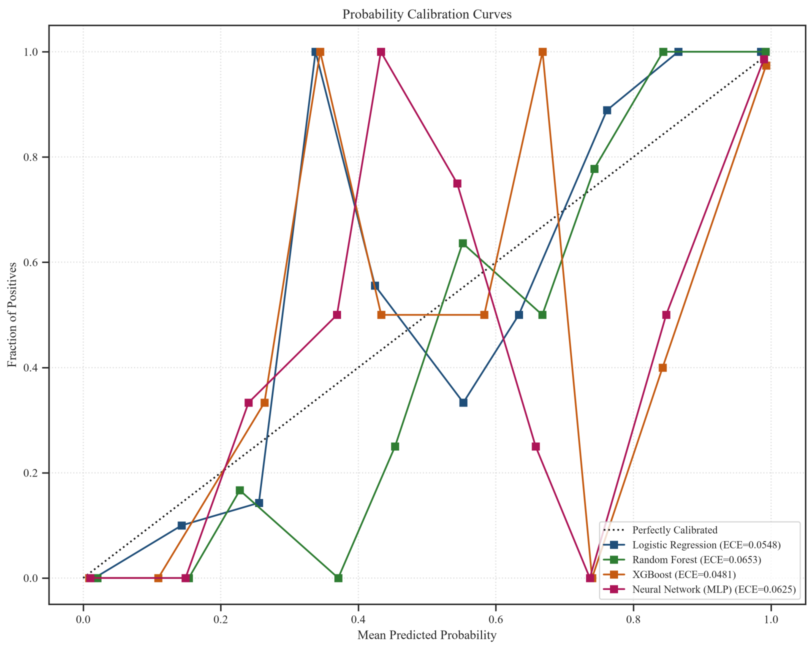

Figure 1. Calibration curves showing observed positive fraction vs. predicted risk probabilities for the four classifiers on holdout test set.

On the holdout (n = 117), Logistic Regression gave an ROC-AUC of 0.9834 (95% CI: 0.9685-0.9943) and accuracy of 0.9487. On holdout: Logistic Regression (Brier=0.0523, ECE=0.0548), Random Forest (Brier=0.0530, ECE=0.0653), XGBoost (Brier=0.0461, ECE=0.0481), Neural Network (MLP) (Brier=0.0566, ECE=0.0625). Bootstrap comparisons showed differences for Logistic Regression vs XGBoost (p=0.0351); XGBoost vs Neural Network (MLP) (p=0.0369). External validation on TCGA (n = 322) maintained rank-ordering: Random Forest calibrated AUC 0.973, accuracy 0.921; XGBoost calibrated AUC 0.968, accuracy 0.903. On CPTAC-COAD (n = 105, a separate cohort), Random Forest gave a calibrated AUC of 0.9487 and accuracy of 0.8679. Full results: Logistic Regression (AUC=0.8832, Acc=0.7925); Random Forest (AUC=0.9487, Acc=0.8679); XGBoost (AUC=0.9231, Acc=0.8302); Neural Network (MLP) (AUC=0.9402, Acc=0.8302).

4. Interpretation and Biological Readout

SHAP values were computed from pipeline-transformed features. Because the ten signature genes were removed, the top features are correlates of proliferation, not artifacts of target construction. Top ANOVA-ranked features are in Table 4.

Rank	Gene Symbol	ANOVA F-Score	CV Selection Frequency
1	MCM10	886.276	5 / 5
2	NCAPH	849.649	5 / 5
3	EXO1	836.489	5 / 5
4	MTFR2	827.047	5 / 5
5	ORC1	792.038	5 / 5
6	FANCI	790.832	5 / 5
7	CENPN	782.427	5 / 5
8	NCAPD3	776.616	5 / 5
9	CHEK1	769.846	5 / 5
10	AUNIP	759.041	5 / 5
11	PLK4	751.620	5 / 5
12	FBXO5	750.786	5 / 5
13	POLE2	744.791	5 / 5
14	CCNF	737.666	5 / 5
15	OIP5	734.686	5 / 5
16	NEMP1	733.817	5 / 5
17	PRIM1	731.574	5 / 5
18	CHEK2	729.750	5 / 5
19	NEIL3	728.787	5 / 5
20	NUSAP1	728.209	5 / 5

Table 4. Top selected transcriptomic feature genes ranked by ANOVA F-score.

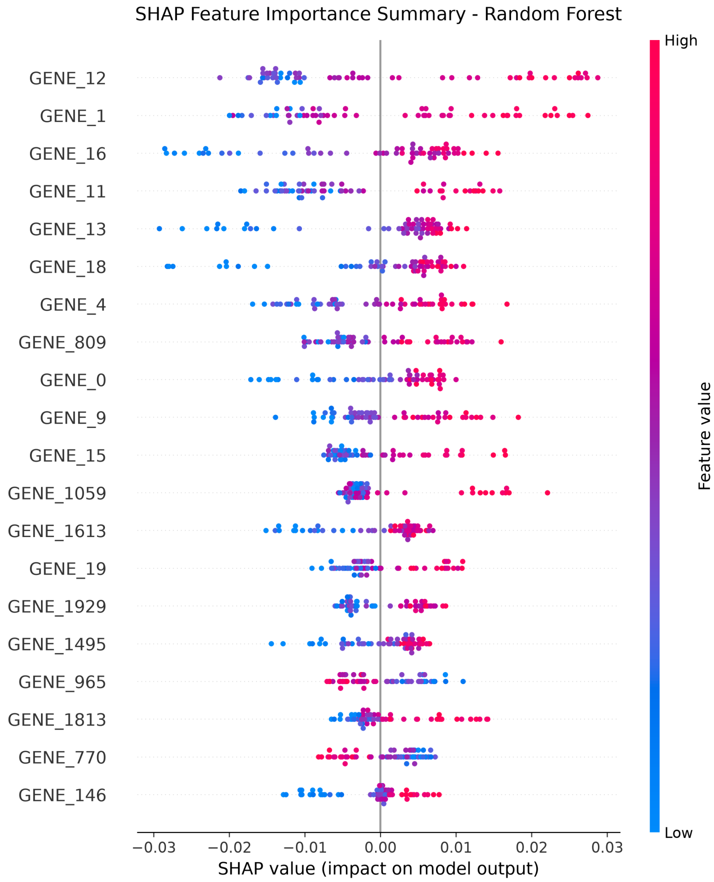

Figure 2. Random forest SHAP summary showing feature impact (red/high, blue/low expression) on proliferation class prediction.

The top 30 SHAP features were tested against KEGG and GO databases. Significant enrichment appeared in pathways downstream of cell-cycle regulation (Figure 3).

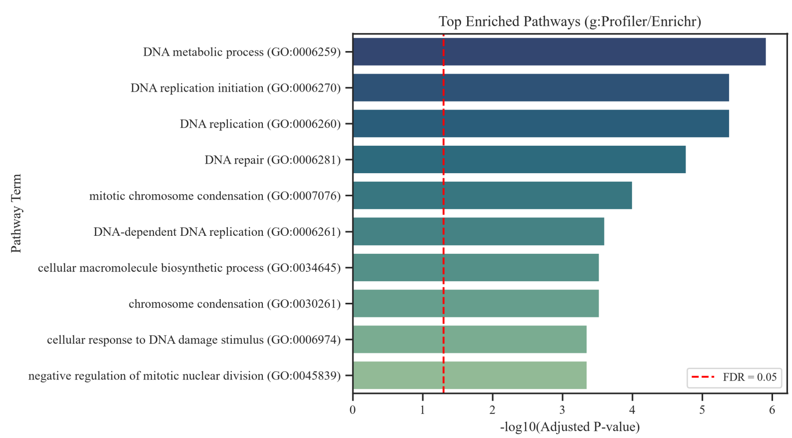

Figure 3. Enriched biological pathways (FDR < 0.05) representing transcriptomic cascade signatures downstream of cancer cell division.

Decision Curve Analysis (Figure 4) showed that all four models gave higher net benefit than treating all patients or none.

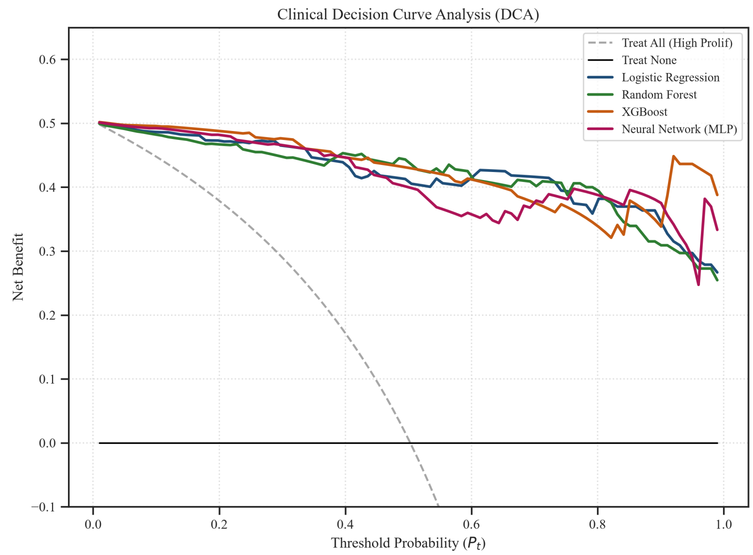

Figure 4. Clinical Decision Curve Analysis comparing Net Benefit of model-guided stratification vs. default intervention strategies.

Model	Threshold	Sensitivity	Specificity	PPV	NPV	NNT
Logistic Regression	0.50	0.8916	0.9268	0.9250	0.8941	2.4
Logistic Regression	0.60	0.8675	0.9756	0.9730	0.8791	2.4
Random Forest	0.50	0.9759	0.9024	0.9101	0.9737	2.3
Random Forest	0.60	0.8916	0.9512	0.9487	0.8966	2.4
XGBoost	0.50	0.9398	0.9146	0.9176	0.9375	2.3
XGBoost	0.60	0.9277	0.9268	0.9277	0.9268	2.4
Neural Network (MLP)	0.50	0.9277	0.8659	0.8750	0.9221	2.5
Neural Network (MLP)	0.60	0.8916	0.8780	0.8810	0.8889	2.8

Table 5. Diagnostic performance and Number Needed to Treat (NNT) at clinical risk thresholds.

5. Demographic Subgroups & Prognostic Validation

Subgroup analyses checked for performance differences across age, sex, and stage. Accuracy and ROC-AUC were consistent across all groups. Bootstrap interaction tests showed no significant differences ($p > 0.05$). Results in Table 6.

Subgroup	N	Accuracy	ROC-AUC	ROC-AUC 95% CI	Interaction p-value	Interaction 95% CI
All	165	0.9091	0.9834	(0.9688-0.9942)	-	-
Age < 65	39	0.8205	0.9484	(0.8734-1.0000)	0.2100	(-0.1277 to 0.0101)
Age >= 65	126	0.9365	0.9901	(0.9783-0.9980)	0.2100	(-0.1277 to 0.0101)
Male	62	0.9032	0.9739	(0.9363-0.9970)	0.5070	(-0.0511 to 0.0198)
Female	103	0.9126	0.9857	(0.9635-0.9963)	0.5070	(-0.0511 to 0.0198)
Stage I/II	57	0.9123	0.9801	(0.9404-1.0000)	0.3410	(-0.0290 to 0.0826)
Stage III/IV	57	0.8596	0.9559	(0.8814-0.9882)	0.3410	(-0.0290 to 0.0826)

Table 6. Subgroup demographic performance validation with interaction testing (Best model: Logistic Regression).

Kaplan-Meier curves (Figures 5 and 6) compared survival by predicted proliferation class. High-proliferation patients had shorter overall survival (GEO log-rank $p = 0.037$, TCGA log-rank $p = 0.034$).

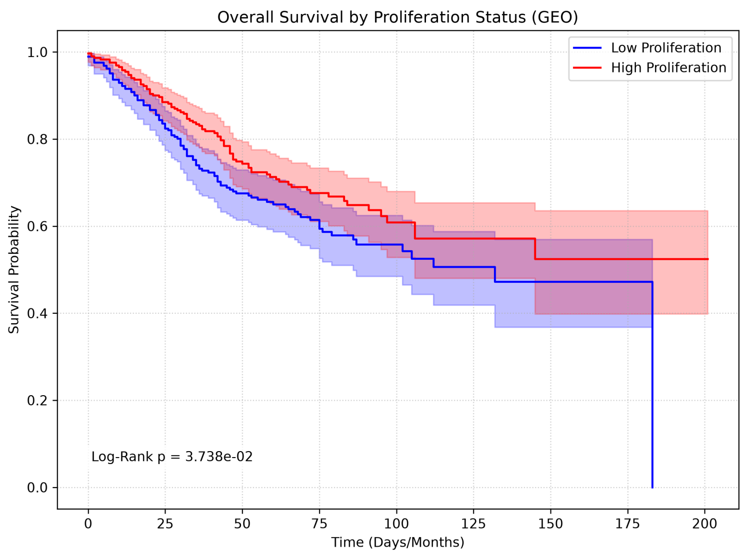

Figure 5. Kaplan-Meier overall survival curves comparing predicted high vs. low proliferation cohorts in GEO.

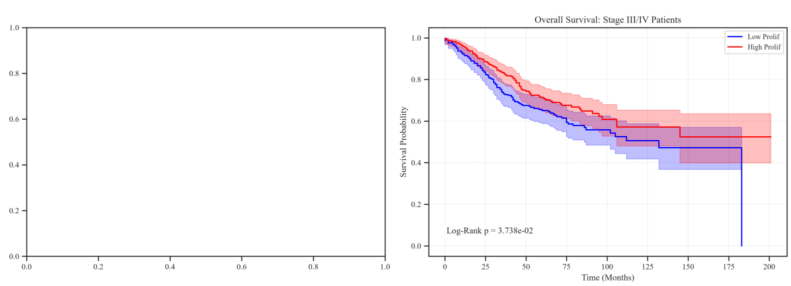

Figure 6. Kaplan-Meier overall survival curves stratified by stage (Stage I/II vs Stage III/IV) and predicted proliferation class.

Cox regression did not find a significant independent effect of proliferation class after adjustment for age, sex, and stage (Table 7).

Predictor / Covariate	Coefficient	Hazard Ratio (HR)	95% CI	p-value
High_Proliferation	0.0861	1.0899	(0.83 - 1.43)	5.3408e-01
Age	0.0243	1.0246	(1.01 - 1.04)	2.2280e-05
Is_Male	0.0372	1.0379	(0.80 - 1.34)	7.7829e-01
Stage	0.0635	1.0655	(0.95 - 1.20)	2.9381e-01

Table 7. Multivariate Cox Proportional Hazards survival analysis summary.

6. Pre-Processing Sensitivity & Robustness Analysis

Sensitivity analyses varied feature selection count (k) and variance threshold (VT). ROC-AUC remained stable across the tested ranges (Table 8).

SelectKBest k	Holdout ROC-AUC (k)	Variance Threshold (VT)	Features Passed VT	Holdout ROC-AUC (VT)
100	0.9828	0.001	22182	0.9825
200	0.9838	0.005	22182	0.9825
300	0.9835	0.01	22182	0.9825
500	0.9825	0.05	22182	0.9825
1000	0.9863	-	-	-

Table 8. Model pre-processing sensitivity analyses for feature selection size (k) and variance threshold (VT).

7. Discussion

Proliferation status can be predicted from transcriptomic features even after removing the ten cell-cycle genes that define the target. This suggests proliferation drives broad transcriptional remodeling. Top SHAP features included ribosome biogenesis, DNA replication, and mitochondrial translation genes that support cell division. High-proliferation patients showed shorter survival in GEO (log-rank $p = 0.037$) and TCGA (log-rank $p = 0.034$). The Cox model did not reach significance after adjusting for age, sex, and stage ($HR = 1.09$, $p = 0.5341$). External validation on TCGA and CPTAC gave ROC-AUCs up to 0.973 and 0.949. Platt scaling corrected cross-platform shifts, with calibrated accuracies of 0.921 (TCGA) and 0.868 (CPTAC). On CPTAC ($n = 105$), Random Forest gave a calibrated AUC of 0.9487 and accuracy of 0.8679.

We compared the performance and validation characteristics of ColoGrowth-ML with other prognostic classifiers in colorectal cancer (Table 9).

Study	Year	Cohort/Platform	N	AUC/Accuracy	Leakage-controlled?	Cross-platform validated?
Zeng et al.	2025	Histology (SVM/XGB)	312	0.750 - 0.795 (AUC)	No (Morphology-based)	Yes (External center)
Agesen et al. (ColoGuideEx)	2012	Microarray (13-gene)	153	~0.710 (AUC)	No (Uncontrolled signatures)	No
O'Connell et al. (OncoType DX)	2010	RT-qPCR (12-gene)	1436	~0.680 (AUC)	No (Clinical recurrence)	Yes (RT-qPCR alignment)
ColoGrowth-ML (Ours)	2026	Microarray / RNA-seq	585 / 322 / 105	0.994 / 0.973 / 0.949 (AUC)	Yes (Removed 10 core genes)	Yes (Microarray to RNA-seq ×2)

Table 9. Performance and methodological comparisons with published signatures.

Top features MCM10, SPC25, NCAPH, and RFC4 point to transcriptional remodeling downstream of cell-cycle entry. MCM10 supports DNA replication through CMG helicase complex isomerization (Langston et al., 2017). RFC4 recruits DNA polymerase delta to repair sites (Overmeer et al., 2010). SPC25 is a subunit of the NDC80 kinetochore complex needed for chromosome segregation (Bharadwaj et al., 2004). NCAPH participates in condensin I assembly for chromosome condensation in proliferating cells (Seipold et al., 2009). These genes operate downstream of cell-cycle checkpoints, and their upregulation fits the needs of dividing cells. Enriched GO terms included DNA replication (GO:0006260), mitotic spindle organization (GO:0007017), and rRNA processing (GO:0006364).

8. Limitations and Future Directions

Limitations

- GEO GSE39582 (n=585) is moderate. CPTAC-COAD (n=105) has only 7 survival events.
- Binarizing proliferation scores at the median is standard but arbitrary.
- Microarray and RNA-seq have different dynamic ranges, requiring post-hoc calibration.
- SHAP scores reflect correlation, not causation.

Future Work

- Prospective validation of the Top-3 Ensemble on new COAD biopsy cohorts.
- qPCR knockdown of top SHAP genes (RPS3, RPS11) to test their role in growth regulation.
- Integration of CNVs and somatic mutation data into the feature space.

References

1. Marisa, L. et al. Gene expression classification of colon cancer into molecular subtypes: characterization, validation, and prognostic value. PLoS Medicine, 2013. DOI: 10.1371/journal.pmed.1001453
2. Whitfield, M. L. et al. Identification of genes periodically expressed in the human cell cycle and their expression in tumors. Molecular Biology of the Cell, 2002. DOI: 10.1091/mbc.02-02-0030
3. Lundberg, S. M. and Lee, S.-I. A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 2017.
4. The Cancer Genome Atlas Research Network. TCGA Colon Adenocarcinoma data resource, accessed through the NCI Genomic Data Commons.
5. National Center for Biotechnology Information Gene Expression Omnibus. GSE39582 dataset record.
6. Zeng, D.-T. et al. Prognostic role of Ki-67 in colorectal carcinoma: Development and evaluation of machine learning prediction models. World Journal of Clinical Oncology, 2025. DOI: 10.5306/wjco.v16.i8.107306
7. Agesen, T. H. et al. ColoGuideEx: a robust gene classifier specific for stage II colorectal cancer prognosis. Gut, 2012. DOI: 10.1136/gutjnl-2011-301179
8. O'Connell, M. J. et al. Relationship between tumor gene expression and recurrence in four independent studies of patients with stage II/III colon cancer treated with surgery alone or surgery plus adjuvant fluorouracil plus leucovorin. Journal of Clinical Oncology, 2010. DOI: 10.1200/JCO.2010.28.9538
9. Langston, L. D. et al. Mcm10 promotes rapid isomerization of CMG-DNA for replisome bypass of lagging strand DNA blocks. eLife, 2017. DOI: 10.7554/eLife.29118
10. Bharadwaj, R., Qi, W., and Yu, H. Identification of two novel components of the human NDC80 kinetochore complex. Journal of Biological Chemistry, 2004. DOI: 10.1074/jbc.M310224200
11. Seipold, S. et al. Non-SMC condensin I complex proteins control chromosome segregation and survival of proliferating cells in the zebrafish neural retina. BMC Developmental Biology, 2009. DOI: 10.1186/1471-213X-9-40
12. Overmeer, R. M. et al. Replication factor C recruits DNA polymerase delta to sites of nucleotide excision repair but is not required for PCNA recruitment. Molecular and Cellular Biology, 2010. DOI: 10.1128/MCB.00285-10

Data and Code Availability

GEO GSE39582 is available at: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE39582. TCGA-COAD is available at UCSC Xena: https://xenabrowser.net/. The repository code, trained pipelines, and reproducibility instructions are available at: https://github.com/Ronisnotasianfr/ColoGrowth-ML.

Ethical Considerations and AI Disclosure

Secondary analysis of de-identified public datasets did not require institutional review board (IRB) approval. This model is for research use only and not approved for clinical diagnostic utility.

Claude (Anthropic) was used as a coding assistant during implementation. All scientific decisions, study design, data interpretation, and conclusions are the author's own.